

SRS - Sprint 1

Software Requirements Specification (SRS) – Recommender System

1. Introduction

This document defines the requirements for a recommender system designed to suggest products based on user behavior data.

The system processes user interactions and generates personalized recommendations using multiple models.

2. System Overview

The system consists of:

- Data Processing (ETL Pipeline)
- Recommendation Models
- Evaluation Module

It operates entirely on local data without external dependencies.

3. Data Requirements

3.1 Input Data

- products.csv / products.parquet
- user_behavior.csv

3.2 Processed Data

- products_clean.parquet
- user_behavior_clean.parquet
- user_features.parquet

4. Functional Requirements

4.1 ETL Pipeline

- Read raw datasets
- Clean missing and duplicate values
- Convert timestamps
- Generate RFM features:
 - Recency
 - Frequency
 - Monetary

4.2 User-Item Matrix

- Construct matrix using purchase interactions
- Rows: users
- Columns: products
- Fill missing values with 0

4.3 Recommendation Models

- Popularity-Based Model
- User-Based Collaborative Filtering
- Item-Based Collaborative Filtering
- Matrix Factorization using SVD

4.4 Evaluation

- Precision@K implementation
- Basic evaluation reporting

5. Non-Functional Requirements

- Runs locally without external APIs
- Minimal dependencies (pandas, sklearn)
- Efficient processing for medium-sized datasets

6. System Constraints

- No real-time data streaming
- No GPU dependency
- Limited to offline batch processing

7. Assumptions

- User behavior data is available and clean
- Product IDs are consistent across datasets
- Interaction data reflects user preferences

8. Future Enhancements

- Hybrid recommendation system
- Integration with APIs for real-time use
- Incorporation of content-based features
- Scalable deployment architecture